

Answers to the referee report''

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The authors would like to thank the referee for the careful reading of the paper and the valuable comments. We have revised the paper according to the referee's report.

Answers to the six comments:

1. page 3, paragraph following Definition 1.1: $L(u(t, \cdot; \mu)$ is missing a parenthesis on the right

A: We have added the missing right parenthesis.

2. page 5, the reviewer finds it instructive to view "Dalang's function" $\Upsilon(\beta)$, for $\beta > 0$, as being an interpolation between its strengthened counterparts, namely:

$$\Upsilon(\beta) \leq 2\Upsilon_\alpha(\beta)^{1/2}\Upsilon_\alpha(0)^{1/2}$$

which can be proven by simply splitting the integral into near and far fields, then optimizing the constant. This would complement the discussion on page 5/6, as well as the one in Section 3.2

A: ...

3. page 7, proof of Corollary 1.2, second equation line: there is a $*$ missing on ζ inside the last integral preceding the inequality. The integrand should be $(G(t, \cdot) * \rho)(y)|\zeta^*(y)|^2$

A: It has been modified as suggested.

4. page 9, definition of H : the authors should include the definition of H since it is invoked several times in the paper and its properties rely on the restrictions associated to the Dalang-type condition. The reviewer went to reference [3] and does not think it will be difficult to add and will only add clarity to their framework.

A: TBD

- 5.

A: We have unified the notation for integral throughout the paper to the convention to be $\int \int dx dy f(x, y)$.

6. page 26, sentence preceding Section 5.4: “Combing” should be “Combining”

A: It has been modified as suggested.

Additional changes that we have made

- 1.
2. We have updated the acknowledgments to include the support from the first author and to express our gratitude to the referee.